# MODULE 2 — AI APPLICATIONS IN HEALTHCARE

Simple, Short & Exam-Oriented Notes

Module focus: Medical imaging + diagnosis + outbreak prediction + Reinforcement Learning (RL) + AI design and deployment.

## MODULE MAP

2.1 Medical Imaging & AI Applications

2.2 Reinforcement Learning in Treatment Planning, Design & Deployment

Exam trick: For application questions, write: Definition → Steps/flow → AI technique → Output → Healthcare use → Safety/limitation.

# 2.1 INTRODUCTION TO MEDICAL IMAGING

Medical imaging means producing visual information about the internal structures or functions of the human body.

In healthcare it supports screening, diagnosis, treatment planning, monitoring and follow-up.

AI/ML can process many images, detect patterns, segment abnormal regions and produce measurements.

## 2.1.1 Medical Imaging and Radiology

Radiology is the clinical field where medical images are acquired, interpreted and used for patient care.

> AI radiology flow: Image acquisition → Quality check → Preprocessing → Image analysis → Output

Possible outputs: disease class, probability/risk score, lesion location or pixel-level segmentation mask.

## Medical Image Processing Stages

| Stage | Purpose | AI/ML role |
|---|---|---|
| Acquisition | Obtain image | Image quality assessment |
| Preprocessing | Standardize image quality | Denoising, normalization, contrast enhancement |
| Segmentation | Separate organ/lesion | Tumor/organ/lesion mask |
| Feature learning | Learn useful visual patterns | CNN/deep features |
| Prediction | Classify/estimate risk | Classification/regression |
| Decision support | Help clinical interpretation | Alerts, prioritization, visualization |

## 2.1.2 Major Medical Imaging Modalities □

| Modality | Simple principle | Main use |
|---|---|---|
| X-ray | Projection image using differential attenuation | Bones, chest and other appropriate examinations |
| CT | X-ray images reconstructed into cross-sectional slices | Tumor location, size, extent and anatomy |
| MRI | Magnetic resonance with strong soft-tissue contrast | Brain, spinal cord and soft tissues |
| PET | Tracer uptake gives metabolic information | Metabolic activity, active cancer regions and disease spread |

> Easy memory: X-ray = projection | CT = cross-section | MRI = soft tissue | PET = metabolic information

> Important: X-ray, CT, MRI and PET are complementary, not interchangeable. Integrated diagnosis can combine anatomical and functional information.

## 2.1.3 Medical Image Processing for Cancer Diagnosis

Cancer diagnosis may combine image processing, ML/DL and clinical information.

The objective may be to detect an abnormality, locate a lesion, segment a tumor or classify disease severity.

Common processing: resizing, intensity normalization, noise reduction, contrast enhancement, segmentation, feature extraction and data augmentation.

| Task | Output | Example |
|---|---|---|
| Classification | Class/probability | Cancerous vs non-cancerous |
| Detection | Location | Locate suspicious lesion |

| Task | Output | Example |
| --- | --- | --- |
| Segmentation | Pixel/voxel mask | Brain tumor boundary |
| Quantification | Measurement | Tumor size/area/volume |

> Memory: Classification = WHAT? | Detection = WHERE? | Segmentation = EXACT REGION? | Quantification = HOW MUCH?

# 2.1.4 DIABETIC RETINOPATHY (DR) DETECTION □□□

DR detection is an image-based screening application. Retinal photographs are analyzed for visual patterns associated with diabetic eye disease.

Typical pipeline:

> Retinal image → Preprocessing → Segmentation/Feature extraction → CNN → Detection/Severity → Screening

## Steps

• Retinal image acquisition: obtain a suitable fundus/retinal image.

• Preprocessing: reduce noise and improve contrast.

• Segmentation/feature extraction: identify useful structures and abnormalities such as blood-vessel changes, microaneurysms and hemorrhages.

• Modeling: train ML/DL, especially CNN, on labeled retinal images.

• Decision: predict DR presence and, if supported by the dataset, classify severity.

• Clinical use: support screening and identify patients needing further ophthalmologist examination.

> Important: AI assists screening; it does not replace clinical examination.

# 2.1.5 BRAIN TUMOR SEGMENTATION IN MRI □□□

The goal is to identify the exact region occupied by the suspected tumor. Therefore the output is a segmentation mask, not just a class label.

> Flow: MRI → Preprocessing → U-Net encoder → bottleneck → decoder → Tumor mask

## U-Net

• Encoder: learns useful image/tissue features.

• Bottleneck: stores high-level/context information.

• Decoder: reconstructs spatial detail and boundaries.

• Output: pixel-level tumor mask.

## Segmentation Metrics

| Metric | Formula | Meaning |
|---|---|---|
| Dice | 2TP/(2TP+FP+FN) | Overlap between predicted and true mask |
| IoU | TP/(TP+FP+FN) | Intersection relative to union |
| Sensitivity | TP/(TP+FN) | Tumor pixels correctly detected |
| Specificity | TN/(TN+FP) | Non-tumor pixels correctly identified |

> Remember: Brain tumor → MRI → U-Net → tumor mask → Dice/IoU

## 2.1.6 MULTIAGENT INFECTIOUS DISEASE PROPAGATION & OUTBREAK PREDICTION

A multiagent model represents individuals or groups as agents. Their health states and interactions change over time.

| Element | Simple meaning |
|---|---|
| Agents | Individuals, households or locations |
| State | Health/infection status |
| Interactions | Contacts, travel, mobility, community mixing |
| Transition | Change in disease state |
| Prediction | Estimate future spread/hotspots/risk |
| Intervention | Isolation, vaccination, resource allocation |

> Flow: Agents + initial states → contacts/mobility → disease-state changes → simulation/ML → forecast → intervention

Useful for understanding disease spread and supporting public-health planning/resource allocation.

Limitation: Forecast quality depends on the quality/representativeness of data and assumptions.

## 2.1.7 AUTOMATED AMBLYOPIA SCREENING SYSTEM

The system aims to identify children who may be at risk of amblyopia so they can be referred for clinical evaluation.

> Flow: Screening information → Quality check → Preprocessing → Feature learning → AI classifier → Risk output → Referral

- Capture suitable visual/ocular screening information.
- Check quality and preprocess it.
- Extract visual/ocular patterns.
- Use an AI classifier to estimate screening risk.
- Generate screen-positive/screen-negative output.
- Refer screen-positive cases for clinical follow-up.

> Important: It is a screening aid. Clinical professionals confirm the condition and manage the patient.

## 2.1.8 INTEGRATED AI-BASED DIAGNOSTIC WORKFLOW

A complete healthcare AI system can combine imaging with patient history, laboratory measurements and other clinical information.

| Function | AI task | Output |
|---|---|---|
| Screening | Normal/abnormal classification | Risk score/class |
| Detection | Find suspicious region | Location/bounding box |
| Segmentation | Delineate lesion | Pixel/voxel mask |
| Diagnosis support | Combine evidence | Probability/decision message |
| Monitoring | Compare serial studies/data | Change estimate/alert |

# 2.2 REINFORCEMENT LEARNING IN TREATMENT PLANNING

Reinforcement Learning (RL) is a framework where an agent interacts with an environment and learns a policy from feedback.

In healthcare: environment = patient or validated simulation; action = treatment decision; feedback = clinical consequence.

## 2.2.1 RL Fundamentals □□□

| RL component | Healthcare meaning | Example |
|---|---|---|
| State | Current patient condition/history | Vitals, labs, symptoms, previous treatment |
| Action | Permitted treatment decision | Select therapy or adjust dosage |
| Reward/Penalty | Feedback from outcome | Improvement = positive; poor/adverse outcome = negative |
| Policy | Strategy mapping states to actions | Treatment decision strategy |
| Environment | Patient or validated simulation | Patient response dynamics |
| Transition | State change after action | New patient state after treatment |

RL loop: State → Action → Patient response → Reward/Penalty → Policy update → New state

Safety: Healthcare actions must follow clinical guidelines, safety rules and physician oversight.

## 2.2.2 RL for Treatment Planning □□□

Treatment planning is a sequential decision problem: the best action now can depend on patient history and how earlier actions affected the patient.

- Represent patient condition as the state.
- Select one permitted treatment action.
- Monitor patient response.
- Good response gives positive feedback; poor/undesirable response gives negative feedback.
- Update the policy so future decisions are better aligned with long-term outcomes.

Goal: Learn a treatment policy that seeks desirable long-term outcomes while respecting clinical constraints.

## 2.2.3 RL Scenario — Medication Adjustment □□□

Within a clinically defined safe action space, the agent can consider maintaining, reducing or increasing a dose.

| Step | Medication example |
|---|---|
| State | Current condition + previous dose-response history |
| Action | Maintain / reduce / increase dose within safe limits |
| Response | Improvement / inadequate response / adverse effect |
| Reward | Favorable outcome without unacceptable adverse effects |
| Penalty | Poor or undesirable/adverse outcome |
| Policy update | Use feedback to refine future decisions |

Very important: The agent must operate inside predefined safe treatment limits. Recommendations require clinical supervision.

## 2.2.4 HEALTHCARE AI DESIGN CONSIDERATIONS □□□

| Consideration | Why important | Safeguard |
|---|---|---|
| Clinical objective | Must optimize a meaningful patient outcome | Define measurable clinical outcome/reward |
| Safety | Some actions may harm patients | Hard action limits, contraindication rules, fail-safe logic |
| Data quality | Bad/incomplete data can mislead | Validation, missing-data handling, quality checks |
| Generalization | Patients/hospitals differ | External validation, representative data |
| Interpretability | Clinicians need to understand behavior | Explanations, confidence, traceable records |
| Human oversight | Clinical responsibility remains essential | Clinician review and override |
| Monitoring | Model behavior can change after deployment | Performance, drift and outcome monitoring |

> Memory: Clinical objective + Safety + Data quality + Generalization + Interpretability + Human oversight + Monitoring

## 2.2.5 DEPLOYMENT OF AI/RL SYSTEMS □□□

High development accuracy alone does not mean a healthcare AI system is ready for clinical use.

> Deployment flow: Offline development → Simulation/retrospective evaluation → Safety validation → Clinical integration → Prospective monitoring → Controlled improvement

| Stage | What happens |
|---|---|
| Offline development | Collect, clean, label, split data; train and validate. |
| Simulation/retrospective | Test on historical or simulated cases. |
| Safety validation | Define permitted actions, limits, thresholds, escalation and fail-safe behavior. |
| Clinical integration | Connect to hospital systems and provide understandable interface. |
| Prospective monitoring | Track outcomes, calibration, drift, false alarms, missed cases and feedback. |
| Continuous improvement | Update only through controlled validation and change management. |

> Deployment principle: Healthcare AI should be treated as a clinical decision-support component with explicit safety boundaries, monitoring and human oversight.

## 2.2.6 PATIENT DETERIORATION PREDICTION □□□

The system processes continuously updated clinical information and converts model output into an actionable alert.

Typical inputs: heart rate, blood pressure, oxygen saturation, temperature, laboratory results and clinical history.

> Flow: Data acquisition → Preprocessing → Risk prediction → Alert threshold → Clinical review → Action/feedback

| Component | Role |
|---|---|
| Data acquisition | Collect vitals, labs and clinical history. |
| Preprocessing | Handle missing values, noise and inconsistent measurements. |
| Prediction model | Estimate probability/risk of deterioration. |

| Component | Role |
| --- | --- |
| Alert module | Compare risk with threshold and generate alert. |
| Clinical review | Healthcare professional assesses alert and acts in context. |
| Feedback | Compare predictions with actual outcomes and improve system. |

> Important: The alert supports timely clinical attention; final interpretation and action remain with healthcare professionals.

# FINAL QUICK REVISION

## Medical Imaging

> X-ray = projection | CT = cross-sectional anatomy | MRI = soft tissue | PET = metabolic information

## Image Tasks

> Classification = class/probability | Detection = location | Segmentation = pixel/voxel mask | Quantification = size/area/volume

## Important Applications

| Application | Remember |
|---|---|
| Cancer imaging | X-ray/CT/MRI/PET + AI → detect/characterize abnormality |
| Diabetic retinopathy | Retinal image → preprocessing → CNN → detection/severity → screening |
| Brain tumor | MRI → U-Net → tumor mask → Dice/IoU |
| Outbreak prediction | Agents + states + contacts/mobility → propagation → forecast → intervention |
| Amblyopia | Screening data → AI risk → referral/clinical confirmation |

## Reinforcement Learning

> State → Action → Response → Reward/Penalty → Policy update → New state

| RL term | Healthcare meaning |
|---|---|
| State | Patient condition/history |
| Action | Treatment/dose decision |
| Reward | Good clinical outcome |
| Penalty | Poor/adverse outcome |
| Policy | Strategy for choosing actions |

## Deployment

> Develop → Simulate → Safety validate → Integrate → Monitor → Controlled improvement

## What to draw in the exam

- Medical image-processing pipeline
- Diabetic retinopathy detection pipeline
- U-Net / brain tumor segmentation flow
- RL treatment-planning loop
- Patient deterioration prediction + alert workflow

# One-Minute Memory Sheet

DR → retinal image + CNN
Tumor → MRI + U-Net + mask + Dice/IoU
Outbreak → agents + interactions + disease states + forecast
Amblyopia → screening + AI risk + referral
RL → state/action/reward/policy
Medication → maintain/reduce/increase safely
Deployment → validate + integrate + monitor + human oversight
Deterioration → vitals/labs → risk → alert → clinician review

## Exam Answer Formula

Definition → 4–6 key points → flow/diagram → healthcare example → safety/limitation.